

CSCI 5612

Machine Learning for Data Science

FORMAT	CATEGORY	CREDITS
On-Campus	Core	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 1 – CS Pathway (on-campus)	Machine Learning Engineer (81.2 / 100)	69.9 / 100
SPECIALIZATION Computer Science Pathway		

70

CURRENT AVG SCORE
across 6 job roles

99

POTENTIAL SCORE
with recommended additions

+37

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Explores the data science lifecycle with a focus on machine learning. Topics include data preparation, unsupervised and supervised analyses, ensemble methods, results illustration, and data communication. Unsupervised methods include clustering, association rule mining, and dimensionality reduction. Supervised models include regression, tree-based models, Bayesian models, and support vector machines. Recommended restrictions: This course is specific to Data Science students and the MS-DS degree program, this course would not be suitable for CSCI majors to meet CS degree requirements.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	81.2	100.0	+18.8	MEDIUM
ML Engineer	77.3	100.0	+22.7	MEDIUM
Data Scientist	70.6	100.0	+29.4	MEDIUM
AI Engineer	67.2	99.1	+31.9	HIGH
MLOps Engineer	63.2	100.0	+36.8	HIGH
Applied Scientist	59.8	94.4	+34.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 5 of 6 roles

systems

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: Feature Engineering — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

81.2 → 100.0 (+18.8 pts)

web sets tools model applications technical systems information

ML Engineer

77.3 → 100.0 (+22.7 pts)

systems model problems product technical building tools

Data Scientist

70.6 → 100.0 (+29.4 pts)

sets product different social problems

AI Engineer

67.2 → 99.1 (+31.9 pts)

following technical model systems problems information product tools

MLOps Engineer

63.2 → 100.0 (+36.8 pts)

model text systems preprocessing tools

Applied Scientist

59.8 → 94.4 (+34.6 pts)

problems technical systems industry regulatory product model techniques

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.